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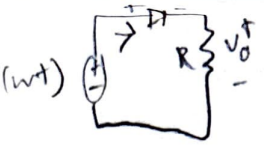
ECEN 164 HW 3

1. (3.1)



a. avg voltage $= \frac{170}{\pi} = 54.113V$

$\frac{54.113}{15} = \boxed{\text{avg curr} = 3.608A}$

$V(t) = 170\sin(377t) \rightarrow 170\angle -90$

$R = 15$

b. $V_{RMS} = \frac{170}{2} = 85V$ (half wave)

$\frac{85}{15} = \boxed{5.667A}$

c. $P = \frac{V_{RMS}^2}{R} = \frac{85^2}{15} =$

$\boxed{481.667W}$

d. $S = V_{s,RMS} I_{s,RMS}$

$\left(\frac{170}{\sqrt{2}}\right)(5.667) = \boxed{681.180VA}$

e. $PF = \frac{481.667}{681.18} = \boxed{\frac{\sqrt{2}}{2}}$

2. (3.19 a+b) 3.51 eq $= \Delta V_o \approx \frac{V_m}{fRC}$

$V_m = 120\sqrt{2}, f = 60$

a.

$\Delta V_o = V_m(1 - \sin\alpha)$

$= V_m(1 - e^{-2\pi/\omega RC})$

abs $\Delta V_o = 120\sqrt{2}(1 - e^{-2\pi/60(1000)(4000\times10^{-6})2\pi}) = \boxed{.706V}$

approx $\Delta V_o = \frac{120\sqrt{2}}{60(4000\times10^{-6})(1000)} = \boxed{\frac{\sqrt{2}}{2}V}$

reasonable ✓

b.

abs $\Delta V_o = 120\sqrt{2}(1 - e^{-2\pi/60(1000)(20\times10^{-6})2\pi}) = \boxed{915.952V}$

approx $\Delta V_o = \frac{120\sqrt{2}}{60(20\times10^{-6})(1000)} = \boxed{100\sqrt{2}V}$

not reasonable

3. (4.5)

$R = 15, L = .03H$

$V_{RMS} = 120V, f = 60$

a. $V_o = \frac{2(120\sqrt{2})}{\pi} = 108.038V$

$Z = 15 + j(60)(.03)$

$Z = 15 + j1.8$

$Z = 15.108\angle 6.843°$

avm, only R used, no non-real part yet.

$\frac{108.038}{15} = \boxed{7.203A}$

b. Harmonics for n=0, 2, 4, 6, 8

$Z_2 = 15 + j(60)(.03)(2)2\pi$ $Z_4 = 15 + j(60)(.03)(4)2\pi$

$= 15 + j22.619$ $= 15 + j45.239$

$|Z_2| = 27.146$ $|Z_4| = 47.661$

$V_2 = 72.025$, $V_4 = 14.405$

$I_{RMS} = \sqrt{7.203^2 + \frac{\left(\frac{72.025}{27.146}\right)^2}{2} + \frac{\left(\frac{14.405}{47.661}\right)^2}{2}}$

$I_{RMS} = 7.446A$

$P = 15(I_{RMS})^2 = \boxed{831.251W}$

c.

$7.446(120) = 893.577VA$

$PF = \frac{831.251}{893.577} = \boxed{.9313}$

4. (4.36) 480V L-L, 60Hz
R = 100Ω, L = 15mH

a. $V_{o(avg)} = \frac{3V_{pm}+\sqrt{2}}{\pi} = 648.228V$

$\frac{648.228}{100} = \boxed{6.483A = I_{o(avg)}}$

n = 6

$V_6 = \frac{6(480\sqrt{2})}{\pi(36-1)}$

$= 37.042V$

n = 12

$V_{12} = \frac{6(480\sqrt{2})}{\pi(144-1)}$

$= 9.066V$

$Z_6 = 100 + j(2\pi 60)(.015)6$

$|Z_6| = 105.50\Omega$

$Z_{12} = 100 + j(2\pi 60)(.015)12$

$|Z_{12}| = 120.85\Omega$

$I_{RMS} \approx \sqrt{6.483^2 + \frac{\left(\frac{37.042}{105.50}\right)^2}{2} + \frac{\left(\frac{9.066}{120.85}\right)^2}{2}}$

$\boxed{I_{RMS} \approx 6.488A}$

Barely bigger

b. $I_{D,avg} = \frac{1}{3} I_{o,avg} = \boxed{2.161A}$

$I_{D,RMS} = \frac{1}{\sqrt{3}} I_{o,RMS} = \boxed{3.7464}$

c. $I_{s,RMS} = \sqrt{\frac{2}{3}} I_{o,RMS} = \boxed{5.297A}$

d. $S = \sqrt{3} V_{L-L,RMS} I_{s,RMS} = \sqrt{3}(480)(5.297) = 4404.250$ VA

$V_{o_{avg}} I_{o_{avg}} = 6.483(648.228) = 4202.462W$

$PF = \frac{4202.462}{4404.250} = \boxed{.954}$

Dylan Thornburg

SANTA CLARA UNIVERSITY	ELEN 164 Winter 2026	S. Krishnan
HW #3		

5. [20 points]

Various designs for multiphase rectifiers are attempted and the outputs of two designs are shown below in Figure 5(a) and 5(b). The y-axis is voltage.

In both figures, the waveforms AC1, AC2 and AC3 are the three phases of the input and DC is the output of the rectifier.

Figure 5(a)

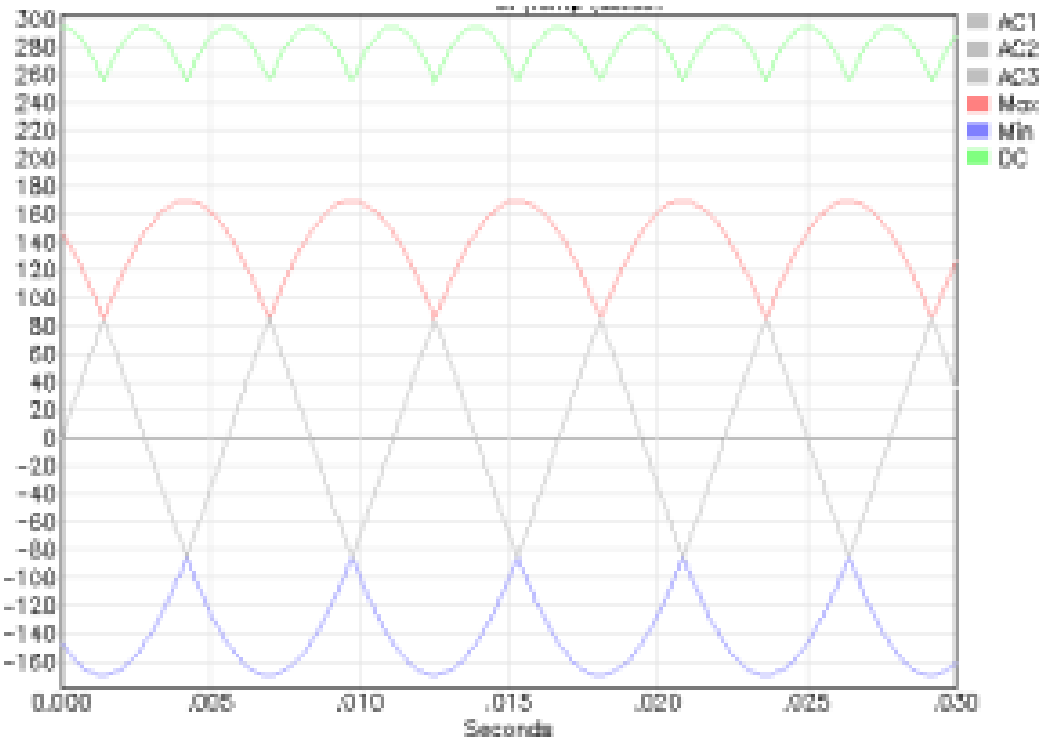
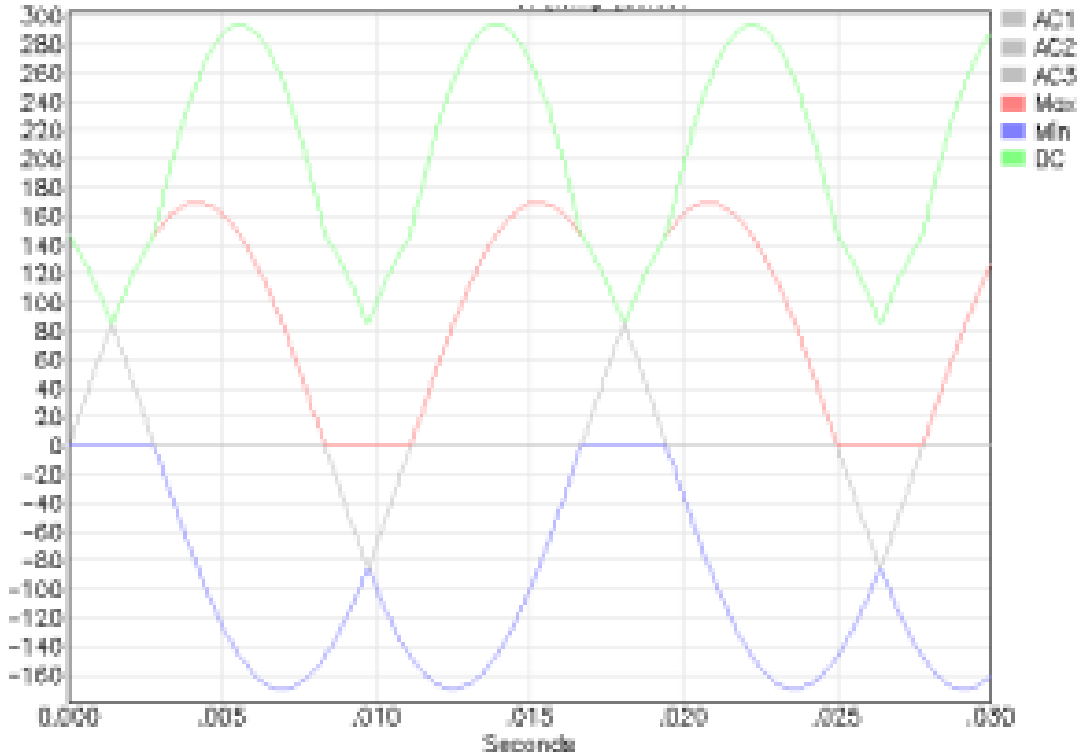


Figure 5(b)



Its overall hard to read these graphs well because of my color blindness, but I'll do my best.

(a) Identify the rectifier that gives the output in Figure 5(a). Explain your answer.
3-phase full bridge rectifier as we get six pulses per AC cycle.

(b) Is the rectifier whose output is shown in Figure 5(b) a better rectifier? Explain why or why not.
NO! Its Vout is far more inconsistent and avg DC value is lower.

(c) Which rectifier (5a or 5b) has more stringent filter requirements? Explain how you arrived at your conclusion.
5b because the "DC" we get from it looks more like an AC signal and thus will require more complex filtering techniques to truly convert to DC. It is also a lower frequency, which is harder to filter.

(d) Why would you want a filter at the output of a rectifier?
Reduce ripple, and furnish it into a steady DC voltage.